

- two polar nuclei during double fertilization. The endosperm provides nourishment to the developing embryo in angiosperm seeds.
- endospore** A thick-coated, resistant cell produced by some bacterial cells when they are exposed to harsh conditions.
- endosymbiont theory** The theory that mitochondria and plastids originated as prokaryotic cells engulfed by a host cell. The engulfed cell and its host cell then evolved into a single organism. *See also* endosymbiosis.
- endosymbiosis** A relationship between two species in which one organism lives inside the cell or cells of another organism. *See also* endosymbiont theory.
- endothelium** (en'-dō-thē'-lē-um) The simple squamous layer of cells lining the lumen of blood vessels.
- endothermic** Referring to organisms that are warmed by heat generated by their own metabolism. This heat usually maintains a relatively stable body temperature higher than that of the external environment.
- endotoxin** A toxic component of the outer membrane of certain gram-negative bacteria that is released only when the bacteria die.
- energetic hypothesis** The concept that the length of a food chain is limited by the inefficiency of energy transfer along the chain.
- energy** The capacity to cause change, especially to do work (to move matter against an opposing force).
- energy coupling** In cellular metabolism, the use of energy released from an exergonic reaction to drive an endergonic reaction.
- enhancer** A segment of eukaryotic DNA containing multiple control elements, usually located far from the gene whose transcription it regulates.
- enteric nervous system** Within the autonomic nervous system, a distinct network of neurons that exerts partially independent control over the digestive tract, pancreas, and gallbladder.
- entropy** A measure of molecular disorder, or randomness.
- enzyme** (en'-zīm) A macromolecule serving as a catalyst, a chemical agent that increases the rate of a reaction without being consumed by the reaction. Most enzymes are proteins.
- enzyme-substrate complex** (en'-zīm) A temporary complex formed when an enzyme binds to its substrate molecule(s).
- eosinophil** Immune system cell that secretes destructive enzymes and helps defend against multicellular pathogens.
- epicotyl** (ep'-uh-kot'-ul) In an angiosperm embryo, the embryonic axis above the point of attachment of the cotyledon(s) and below the first pair of miniature leaves.
- epidemic** A widespread outbreak of a disease.
- epidermis** (1) The dermal tissue of nonwoody plants, usually consisting of a single layer of tightly packed cells. (2) The outermost layer of cells in an animal.
- epididymis** (ep'-uh-did'-uh-mus) A coiled tubule located adjacent to the mammalian testis where sperm are stored.
- epigenetics** The study of the inheritance of traits transmitted by mechanisms that do not involve the nucleotide sequence.
- epinephrine** (ep'-i-nef'-rin) A catecholamine that, when secreted as a hormone by the adrenal medulla, mediates “fight-or-flight” responses to short-term stresses; also released by some neurons as a neurotransmitter; also called adrenaline.
- epiphyte** (ep'-uh-fit) A plant that nourishes itself but grows on the surface of another plant for support, usually on the branches or trunks of trees.
- epistasis** (ep'-i-stā'-sis) A type of gene interaction in which the phenotypic expression of one gene alters that of another independently inherited gene.
- epithelial tissue** (ep'-uh-thē'-lē-ul) Sheets of tightly packed cells that line organs and body cavities as well as external surfaces.
- epithelium** (plural, **epithelia**) An epithelial tissue.
- epitope** A small, accessible region of an antigen to which an antigen receptor or antibody binds.
- equilibrium potential ( $E_{ion}$ )** The magnitude of a cell's membrane voltage at equilibrium; calculated using the Nernst equation.
- erythrocyte** (eh-rith'-ruh-sīt) A blood cell that contains hemoglobin, which transports oxygen; also called a red blood cell.
- erythropoietin (EPO)** (eh-rith'-rō-poy'-uh-tin) A hormone that stimulates the production of erythrocytes. It is secreted by the kidney when body tissues do not receive enough oxygen.
- esophagus** (eh-sof'-uh-gus) A muscular tube that conducts food, by peristalsis, from the pharynx to the stomach.
- essential amino acid** An amino acid that an animal cannot synthesize itself and must be obtained from food in prefabricated form.
- essential element** A chemical element required for an organism to survive, grow, and reproduce.
- essential fatty acid** An unsaturated fatty acid that an animal needs but cannot make.
- essential nutrient** A substance that an organism cannot synthesize from any other material and therefore must absorb in preassembled form.
- estradiol** (es'-truh-dī'-ol) A steroid hormone that stimulates the development and maintenance of the female reproductive system and secondary sex characteristics; the major estrogen in mammals.
- estrogen** (es'-trō-jen) Any steroid hormone, such as estradiol, that stimulates the development and maintenance of the female reproductive system and secondary sex characteristics.
- estrous cycle** (es'-trus) A reproductive cycle characteristic of female mammals except humans and certain other primates, in which the endometrium is reabsorbed in the absence of pregnancy and sexual response occurs only during a mid-cycle point known as estrus.
- estuary** The area where a freshwater stream or river merges with the ocean.
- ethylene** (eth'-uh-lēn) A gaseous plant hormone involved in responses to mechanical stress, programmed cell death, leaf abscission, and fruit ripening.
- etiolation** Plant morphological adaptations for growing in darkness.
- euchromatin** (yū-krō'-muh-tin) The less condensed form of eukaryotic chromatin that is available for transcription.
- eudicot** (yū-dī'-kot) A member of a clade that contains the vast majority of flowering plants that have two embryonic seed leaves, or cotyledons.
- euglenid** (yū'-glen-id) A protist, such as *Euglena* or its relatives, characterized by an anterior pocket from which one or two flagella emerge.
- euglenozoan** A member of a diverse clade of flagellated protists that includes predatory heterotrophs, photosynthetic autotrophs, and pathogenic parasites.
- Eukarya** (yū-kār'-ē-uh) The domain that includes all eukaryotic organisms.
- eukaryote** A single-celled or multicellular organism comprised of eukaryotic cells; eukaryotes include protists, plants, fungi, and animals.
- eukaryotic cell** (yū'-kār-ē-ot'-ik) A type of cell with a membrane-enclosed nucleus and membrane-enclosed organelles. Organisms with eukaryotic cells (protists, plants, fungi, and animals) are called eukaryotes.
- eumetazoan** (yū'-met-uh-zō'-un) A member of a clade of animals with true tissues. All animals except sponges and a few other groups are eumetazoans.
- eurypterid** (yur-ip'-tuh-rid) An extinct carnivorous chelicerate; also called a water scorpion.
- Eustachian tube** (yū-stā'-shun) The tube that connects the middle ear to the pharynx.
- eutherian** (yū-thēr'-ē-un) Placental mammal; mammal whose young complete their embryonic development within the uterus, joined to the mother by the placenta.
- eutrophic lake** (yū-trōf'-ik) A lake that has a high rate of biological productivity supported by a high rate of nutrient cycling.
- eutrophication** A process by which nutrients, particularly phosphorus and nitrogen, become highly concentrated in a body of water, leading to increased growth of organisms such as algae or cyanobacteria.
- evaporative cooling** The process in which the surface of an object becomes cooler during evaporation, a result of the molecules with the greatest kinetic energy changing from the liquid to the gaseous state.
- evapotranspiration** The total evaporation of water from an ecosystem, including water transpired by plants and evaporated from a landscape, usually measured in millimeters and estimated for a year.
- evo-devo** Evolutionary developmental biology; a field of biology that compares developmental processes of different multicellular organisms to understand how these processes have evolved and how changes can modify existing organismal features or lead to new ones.
- evolution** Descent with modification; the process by which species accumulate differences from their ancestors as they adapt to different environments over time; also defined as a change in the genetic composition of a population from generation to generation.
- evolutionary lineage** The sequence of ancestral organisms leading to a particular taxon;